5. Free Chat

1. Dialogue Interaction Flow

Each dialogue of AI Xiaozhi follows this lifecycle:

```
┌──────────┐
│  IDLE    │  Device idle, waiting for wake-up
│  Idle standby │  LED breathing light / Display shows default expression
└────┬─────┘
     │ Hear "Hi,Andy" (offline wake word detection)
     ▼
┌──────────┐
│LISTENING │  Device enters audio pickup state
│  Listening │  LED lights up / Screen shows "Listening..."
└────┬─────┘
     │ VAD detects speech end / User manually stops
     ▼
┌──────────┐
│CONNECTING│  Send recording data, wait for AI reply
│  Thinking │  Screen shows "Other party is speaking..." / animation
└────┬─────┘
     │ Server returns TTS audio
     ▼
┌──────────┐
│SPEAKING  │  Play AI reply voice
│  Speaking │  LED bounces with speech / Screen shows reply text
└────┬─────┘
     │ Playback complete
     ▼
┌──────────┐
│  IDLE    │  Return to idle, wait for next wake-up
└──────────┘
```

Three Listening Modes:

Mode	Behavior	Applicable Scenario
`AutoStop` (default)	Auto-detect speech end, auto-send after 1.5s silence	Daily conversation
`ManualStop`	User manually presses button to end recording	Long recording / noisy environment
`Realtime`	Real-time streaming transmission (requires AEC)	Ultra-low latency scenarios

2. Wake Word Interaction

Wake Device

Speak the default wake word **"Hi,Andy"** (or your custom wake word) toward the device; the device will:

1. **Sound feedback**: Emit a short "ding" sound
2. **Visual feedback**: LED lights up / screen shows listening status
3. **State switch**: Enter `Listening` state, start recording

```
// Woken from idle
I (18912) Application: Wake word detected: Hi,Andy (state: 0)
I (18914) StateMachine: State: idle -> listening

// User finished speaking, VAD detects silence, auto-send
I (315705) Application: >> How is the weather today.
I (315723) StateMachine: State: listening -> speaking

// Server returns TTS voice
17:00:22.776 -> I (316266) Application: << % get_weather...
17:00:23.778 -> I (317294) Application: << It's sunny and warm in Shenzhen today,
17:00:26.224 -> I (319720) Application: << with a temperature of 33°C. It feels like 36°C, so it's a bit humid.
17:00:31.884 -> I (325374) SystemInfo: free sram: 86003 minimal sram: 77279
17:00:32.620 -> I (326139) Application: << The wind is from the southeast at a gentle breeze.
17:00:35.990 -> I (329499) Application: << Perfect weather for a sunny day!

// Return to idle after playback
I (41580) StateMachine: State: speaking -> idle
```

```
17:00:22.186 -> I (315705) Application: >> How is the weather today.
17:00:22.225 -> I (315723) StateMachine: State: listening -> speaking
17:00:22.776 -> I (316266) Application: << % get_weather...
17:00:23.778 -> I (317294) Application: << It's sunny and warm in Shenzhen today,
17:00:26.224 -> I (319720) Application: << with a temperature of 33°C. It feels like 36°C, so it's a bit humid.
17:00:31.884 -> I (325374) SystemInfo: free sram: 86003 minimal sram: 77279
17:00:32.620 -> I (326139) Application: << The wind is from the southeast at a gentle breeze.
17:00:35.990 -> I (329499) Application: << Perfect weather for a sunny day!
17:00:38.264 -> I (331783) Application: << Would you like to know the forecast for the rest of the week?
17:00:41.358 -> I (334837) StateMachine: State: speaking -> listening
```

Interrupt Dialogue

During AI broadcast, you can interrupt it anytime by speaking the wake word:

```
// Wake from idle, enter listening
I (13524) Application: Wake word detected: Hi,Andy (state: 0)
I (13540) StateMachine: State: idle -> listening

// User speaks → server replies
I (15057) Application: >> Hi,Andy.
I (15060) StateMachine: State: listening -> speaking
I (15060) Application: << Hi, hello!
I (16888) Application: << What can I help you with today?

// Return to idle after playback
I (16900) StateMachine: State: speaking -> idle

// Wake again, interrupted during broadcast
I (28000) Application: Wake word detected: Hi,Andy (state: 0)
I (28002) StateMachine: State: idle -> listening
I (82904) Application: >> Tell me a story.
I (82944) StateMachine: State: listening -> speaking
I (83511) Application: << Oh, totally!
I (84956) Application: << Once upon a time, in a cozy little village,
I (87664) Application: << there was a tiny blue bird who loved to sing. Every morning, he’d fly to the tallest tree and belt out a cheerful tune.

// User speaks wake word to interrupt playback
I (41356) Application: Wake word detected: Hi,Andy (state: 6)
I (41357) Application: Abort speaking
I (41359) StateMachine: State: speaking -> listening
```

> The interrupt function is implemented through `kAbortReasonWakeWordDetected` mechanism—immediately stop playback queue and clear audio buffer after wake word is detected.

```
17:17:18.393 -> I (82904) Application: >> Tell me a story.
17:17:18.433 -> I (82944) StateMachine: State: listening -> speaking
17:17:19.019 -> I (83511) Application: << Oh, totally!
17:17:20.445 -> I (84956) Application: << Once upon a time, in a cozy little village,
17:17:23.183 -> I (87664) Application: << there was a tiny blue bird who loved to sing. Every morning, he’d fly to the tallest tree and belt out a cheerful tune.
17:17:24.525 -> I (88996) Application: Wake word detected: Hi, Andy(state: 6)
17:17:24.525 -> I (88996) Application: Abort speaking
17:17:24.525 -> I (88998) StateMachine: State: speaking -> listening
```

3. Button Interaction

Button	Current State	Behavior After Press
BOOT (GPIO0)	Idle / Playing (IDLE / SPEAKING)	Enter listening mode, same as saying "Hi,Andy"
BOOT (GPIO0)	Listening (LISTENING)	Exit listening, return to idle standby
BOOT (GPIO0)	Starting (kDeviceStateStarting)	Enter Wi-Fi provisioning mode
RST	Any state	Hardware reset

4. Voice Commands Available During Dialogue

In addition to free chat, Xiaozhi also provides some built-in commands through **MCP tools**:

Voice Command Example	MCP Tool	Effect
"Set volume to 50%"	`set_volume`	Adjust playback volume
"What's your name"	LLM directly answers	"I'm Xiaozhi, your AI assistant"
"Tell a joke"	LLM directly answers	Tell a joke
"What time is it now"	NTP sync	Broadcast current time

5. Common Troubleshooting

Phenomenon	Cause	Troubleshooting Steps
No response when saying wake word	Wake word model not loaded	Check serial logs for "ESP-SR initialized"
Echo/feedback after wake-up	Speaker sound entering MIC	Enable AEC (`CONFIG_USE_DEVICE_AEC`) or lower volume
Dialogue stuck at "Thinking"	Server not responding / timeout	Check network status, view server logs
Incomplete reply voice	TTS output truncated	Check network latency, consider using faster TTS model
Inaccurate speech recognition	Large ambient noise / dialect	Move closer to MIC, reduce ambient noise, check ASR model configuration
Device frequently disconnects	Wi-Fi signal weak	Move closer to router, or configure Wi-Fi reconnection strategy
Reboot caused by insufficient PSRAM	Memory leak or insufficient resources	Check heap memory usage, close unnecessary features
Playback sound choppy	Insufficient audio buffer	Increase `MAX_PLAYBACK_TASKS_IN_QUEUE`

Login to xiaozhi.me console → **Dialogue Records**, you can view complete ASR text, LLM reply, and TTS text for each dialogue, convenient for locating issues.

Appendix A · Hardware Pin Quick Reference

Pin	Macro	Function
GPIO15	`AUDIO_I2S_GPIO_MCLK`	I2S MCLK
GPIO18	`AUDIO_I2S_GPIO_WS`	I2S WS

Pin	Macro	Function
GPIO16	`AUDIO_I2S_GPIO_BCLK`	I2S BCLK
GPIO17	`AUDIO_I2S_GPIO_DIN`	I2S DIN (MIC)
GPIO8	`AUDIO_I2S_GPIO_DOUT`	I2S DOUT (SPK)
GPIO41	`AUDIO_CODEC_I2C_SDA_PIN`	ES8311 I2C SDA
GPIO42	`AUDIO_CODEC_I2C_SCL_PIN`	ES8311 I2C SCL
GPIO7	`AUDIO_CODEC_PA_PIN`	NS4150 PA enable
GPIO48	`BUILTIN_LED_GPIO`	WS2812 RGB LED
GPIO0	`BOOT_BUTTON_GPIO`	BOOT button

Appendix B · Reference Resources

Resource	Link
AI Xiaozhi GitHub Repository	78/xiaozhi-esp32
Official Server Console	xiaozhi.me
Xiaozhi Encyclopedia	Feishu Docs
Beginner Flash Tutorial	Feishu Docs
Python Server	xinnan-tech/xiaozhi-esp32-server
Java Server	joey-zhou/xiaozhi-esp32-server-java
Golang Server	AnimeAIChat/xiaozhi-server-go
Assets Online Generator	78/xiaozhi-assets-generator
ESP-SR Speech Recognition	espressif/esp-sr
Video Tutorial (BiliBili)	Install Camera for Xiaozhi